hw10

Ch 22

Problem 4

(4)

Let α be a zero of x^3+x^2+1 over $\mathbb{Z}_2$

$\alpha^3+\alpha^2+1=0$ so $\alpha^3=\alpha^2+1$

so we have the field $\mathbb{Z}_2(\alpha)$. This field has 8 elem b/c $|\mathbb{Z}_2|=2$ + there are 3 sols to x^3+x^2+1 so $2^3=8$ elem. Elements: $\{0, 1, \alpha, \alpha+1, \alpha^2, \alpha^2+1, \alpha^2+\alpha+1, \alpha^2+\alpha\}$

Know $x+\alpha$ is a root so

$$
\begin{array}{r|l}
 & x^2+x(1+\alpha)+(\alpha^2+\alpha) \\
x+\alpha & x^3+x^2+1 \\
 & x^3+x^2\alpha \\
\hline
 & x^2+x^2\alpha+1 = x^2(1+\alpha)+1 \\
 & x^2(1+\alpha)+x\alpha(1+\alpha) \\
\hline
 & x\alpha(1+\alpha)+1 = x(\alpha^2+\alpha)+1 \\
 & x(\alpha^2+\alpha)+\alpha^3+\alpha^2 \\
\hline
 & \alpha^3+\alpha^2+1
\end{array}
$$

Know $\alpha^3+\alpha^2=1$ so $=1+1=0$

so $r=0$

so $x^3+x^2+1=(x+\alpha)(x^2+x(1+\alpha)+(\alpha^2+\alpha))$

Know if α is a root, then α^p + α^{p^2} are roots where $p=$ size of original field $=|\mathbb{Z}_2|=2$

so α^2 + $\alpha^4=\alpha^3(\alpha)=(\alpha^2+1)(\alpha)=\alpha^3+\alpha=\alpha^2+\alpha+1$

Check: Plug into $x^2+x(1+\alpha)+(\alpha^2+\alpha)$

α^2: $\alpha^4+\alpha^2+\alpha^3+\alpha^2+\alpha=\alpha(\alpha^2+1)+\alpha^2+1+\alpha=\alpha^3+\alpha+\alpha^2+1+\alpha=\alpha^3+\alpha^3=0$ ✓

$\alpha^2+\alpha+1$: $(\alpha^2+\alpha+1)(\alpha^2+\alpha+1)+(\alpha^2+\alpha+1)(1+\alpha)+\alpha^2+\alpha=$

$\alpha^4+\cancel{\alpha^3}+\cancel{\alpha^2}+\cancel{\alpha^3}+\cancel{\alpha^2}+\cancel{\alpha}+\cancel{\alpha^2}+\cancel{\alpha}+1+\cancel{\alpha^2}+\alpha+1+\alpha^3+\cancel{\alpha^2}+\cancel{\alpha}+\cancel{\alpha^2}+\cancel{\alpha}$

$=\alpha^4+\alpha^3+\alpha=\alpha(\alpha^2+1)+\alpha^2+1+\alpha=\alpha^3+\alpha^2+1=0$ ✓

Problem 5

5 $27 = 3^3$ so we are in $\mathbb{Z}_3$ w/ a poly of deg 3 such that the polynomial has no roots.

$x^3 + x^2 + 1$: 1 is a root $x^3 + 1$: 2 is a root

$x^3 + x + 1$: 1 is a root $f(x) = x^3 + x^2 + x + 1$: $f(0) = 1$, $f(1) = 1$, $f(2) = 0$

$x^3 + 2x + 1$: $f(0) = 1$, $f(1) = 1$, $f(2) = 1$ so

$x^3 + 2x + 1$

so then $\mathbb{Z}_3[x]/(x^3 + 2x + 1)$ has 27 elem

$\cong \mathbb{Z}_3(\alpha)$ where α is a root of

$x^3 + 2x + 1$

Problem 7

(7)(1) $f(x) = x^5 - 1 = x^5 + 1$ 1 is a root so

$$x+1 \,\overline{)\, x^5+1} \quad \text{quotient } x^4+x^3+x^2+x+1$$

x^5+x^4

x^4+1

x^4+x^3

x^3+1

x^3+x^2

x^2+1

x^2+x

$x+1$

$$\boxed{f(x) = (x+1)(x^4+x^3+x^2+x+1)}$$

no roots in $\mathbb{Z}_2$

$$(x^2+x+1)(x^2+x+1) = x^4 + \cancel{x^3} + x^2 + \cancel{x^3} + \cancel{x^2} + \cancel{x} + \cancel{x^2} + \cancel{x} + 1 = x^4+x^2+1$$

so $x^4+x^3+x^2+x+1$ is irred in $\mathbb{Z}_2$ so done

all deg 1 poly have a root so if no deg 2 divisors, then irreducible

(2) No roots so check quadratic factors

$$x^2+x+1 \,\overline{)\, x^6+x^5+x^4+x^3+x^2+x+1} \quad \text{quotient } x^4 + x,\ r\ 1$$

so no check cubic

$x^6+x^5+x^4$

$0+x^3+x^2+x$

x^3+x^2+x

$0+1$

$$x^3+x+1 \,\overline{)\, x^6+x^5+x^4+x^3+x^2+x+1} \quad \text{quotient } x^3+x^2+1$$

so

$x^6+x^4+x^3$

x^5+x^2+x

$x^5+x^3+x^2$

x^3+x+1

x^3+x+1

0

$$x^6+x^5+x^4+x^3+x^2+x+1 = (x^3+x^2+1)(x^3+x+1)$$

(3) $x^9-1 = x^9+1 = (x^3+1)^2$ b/c by freshman dream

1 is a root so

$$x+1 \,\overline{)\, x^3+1} \quad \text{quotient } x^2+x+1$$

x^3+x^2

x^2+1

x^2+x

$x+1$

so $x^9-1 = ((x+1)(x^2+x+1))^2$

irred in $\mathbb{Z}_2$

(4) Showed in q1 that $x^4+x^3+x^2+x+1$ is irreducible

Problem 8

(8) Yes. x^3+x+1 is irreducible in $\mathbb{Z}_2$ & has 3 roots
so $|\mathbb{Z}_2[x]/(x^3+x+1)| = 2^3 = 8$. same w/ x^3+x^2+1
so $|\mathbb{Z}_2[x]/(x^3+x^2+1)| = 2^3 = 8$
Know that for any field of ~~power~~ size p^n (p prime),
there is 1 field up to isomorphism so
since both fields have size 2^3, there is
only one field of that size so they are
isomorphic

Ch 23

Problem 1

1. (1) $G(\mathbb{Q}(\sqrt{30})/\mathbb{Q}) = \{\sqrt{30} \to \sqrt{30}, \sqrt{30} \to -\sqrt{30}\} \cong \mathbb{Z}_2$

$|G(\mathbb{Q}(\sqrt{30})/\mathbb{Q})| = [\mathbb{Q}(\sqrt{30}) : \mathbb{Q}] = 2$

Normal b/c for $x^2 - 30$, the roots are $\sqrt{30}, -\sqrt{30} \in \mathbb{Q}(\sqrt{30})$

(2) $\mathbb{Q}(\sqrt[4]{5})$: Only 1 generator $\sqrt[4]{5}$ so $G(\mathbb{Q}(\sqrt[4]{5})/\mathbb{Q}) = \{\sqrt[4]{5} \to \sqrt[4]{5}, \sqrt[4]{5} \to -\sqrt[4]{5}\}$

$x^4 - 5$ has $\sqrt[4]{5}, -\sqrt[4]{5}, \sqrt[4]{5}i, -\sqrt[4]{5}i$ so not normal — 4 elems

b/c $-\sqrt[4]{5}i \notin \mathbb{Q}(\sqrt[4]{5})$. Normal ext: need i so $\mathbb{Q}(\sqrt[4]{5}, i)$

(3) $G(\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})/\mathbb{Q}) = \{\{\alpha, \beta, \gamma\}, \{\alpha, \beta, -\gamma\}, \{\alpha, -\beta, \gamma\}, \{\alpha, -\beta, -\gamma\},$
8 elems
$\{-\alpha, \beta, \gamma\}, \{-\alpha, \beta, -\gamma\}, \{-\alpha, -\beta, \gamma\}, \{-\alpha, -\beta, -\gamma\}\}$

where α is $\sqrt{2} \to \sqrt{2}$, $-\alpha$ is $\sqrt{2} \to -\sqrt{2}$, β is $\sqrt{3} \to \sqrt{3}$, $-\beta$ is $\sqrt{3} \to -\sqrt{3}$

+ γ is $\sqrt{5} \to \sqrt{5}$ + $-\gamma$ is $\sqrt{5} \to -\sqrt{5}$

$(x - \sqrt{2})(x + \sqrt{2})(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{5})(x + \sqrt{5}) = x^6 - 10x^4 + 31x^2 - 30$

all of the roots of this poly are in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ so normal

(4) ~~same galois group as 3 just α is $\sqrt{2} \to \sqrt{2}$,~~
~~β is $\sqrt[3]{2} \to \sqrt[3]{2}$, γ is $i \to i$~~
~~$f(x) = x^6 - 2$~~

Since $\sqrt{2}$ + $\sqrt[3]{2}$ are in the field, $\frac{\sqrt{2}}{\sqrt[3]{2}} = \sqrt[6]{2}$ is in field so

$\sqrt[6]{2}, \sqrt[3]{2}, \sqrt{2}, \sqrt[3]{4}, \sqrt[6]{32}$ are in the field so $\mathbb{Q}(\sqrt[6]{2}, i) =$

$\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}, i)$ so Galois group: $\{e, \{e, i \to -i\}, \{\sqrt[6]{2} \to \sqrt[6]{2}, e\},$

$\{\sqrt[6]{2} \to -\sqrt[6]{2}, i \to -i\}\}$, 4 elem

Know the polynomial is $x^6 - 2$ which has root $\sqrt{3}$

but $\sqrt{3}$ not in the field so not norm

Norm: $\mathbb{Q}(\sqrt[6]{2}, \sqrt{3}, i)$

5: group: $\{\{e, e\}, \{e, i \to -i\}, \{\sqrt{6} \to -\sqrt{6}, e\}, \{\sqrt{6} \to -\sqrt{6}, i \to -i\}\}$
4 elems

$p(x) = (x^2+1)(x^2-6)$ has all 4 roots in $\mathbb{Q}(\sqrt{6}, i)$
so normal

Problem 2

2 1 $x^3+2x^2-x-2 = x^2(x+2)-(x+2) = (x+2)(x^2-1) =$
$(x+2)(x+1)(x-1)$ so yes (b/c all roots in $\mathbb{Q}$

2 $x^4+2x^2+1 = (x^2+1)^2$ roots are $i, -i \notin \mathbb{Q}$ so not seperable

3 -2 is a root so divable by $x+2$ & $x+1$ which means the rest is deg 2.

$x+1 \overline{\big)\, x^4+x^2+1}$, quotient $2x^3+2x^2+x+2$: $2x^4+2x^3$; $2x^3+x^2$; x^3+x^2; $2x^2$; x^2+x; $x+1$; $2x+2$; 0

$x+2 \overline{\big)\, 2x^3+x^2+x+2}$, quotient x^2+2: x^3+2x^2; $0+x+2$; $2x+1$; 0

$x+1 \overline{\big)\, x^2+2}$, quotient $2x^2+1$: $2x^2+2x$; $2x+2$

So $(x+2)(x+1)^2(2x+1)$
not seperable b/c
$(x+1)$ has multiplicity of 2

4 x^3+x^2+1 has no roots so not divable by deg 1. If it is divable by deg 2, then the quotient is deg 1 which is not possible. So not seperable

Problem 4

Note, don't need Solvability acording to prof

① 3 real roots + 2 imaginary Let G denote galois group. Since $f(x)$ is irreducible of deg 5 in $\mathbb{Q}$ so 5-cycle in G. Also 2-cycle in G b/c we have a complex root so b/c 5-cycle + 2-cycle $\in G \leq S_5$, $\boxed{G \cong S_5}$ ~~Not solvable by radicals b/c complex roots~~

2 3 real + 2 imaginary so Galois group $\cong S_5$, ~~not solvable by radicals~~

3 roots: $\sqrt[3]{5}$, $\sqrt[3]{5}w$, $\sqrt[3]{5}w^2$ where $w = e^{2\pi i/3}$. 3-cycle + transposition so S_3

4 $x^4 - x^2 - 6 = (x^2-3)(x^2+2)$ roots: $\pm\sqrt{3}, \pm\sqrt{2}i$ so $\mathbb{Z}_2 \times \mathbb{Z}_2$

5 $(x^5+1) = (x+1)(x^4 - x^3 + x^2 - x + 1)$. No idea according to Wolfram Alpha, the ~~gro~~ galois group $\cong \mathbb{Z}_4$. The real root doesn't affect + the other roots have 4th root so $\mathbb{Z}_4 \times \{1\} = \mathbb{Z}_4$

6 roots: $\pm\sqrt{2}, \pm\sqrt{2}i$ so $\mathbb{Z}_2 \times \mathbb{Z}_2$

7 roots: $\pm i, \pm 1, \pm\frac{1\pm i}{\sqrt{2}}$ so need $\sqrt{2}$ + i so $\mathbb{Q}(\sqrt{2}, i)$ so galois group is $\mathbb{Z}_2 \times \mathbb{Z}_2$

8 roots: $\pm\cos(\frac{\pi}{8}) \pm i\sin(\frac{\pi}{8})$, $\pm\sin(\frac{\pi}{8}) \pm i\cos(\frac{\pi}{8})$ so $\mathbb{Q}\left(\frac{\sqrt{2+\sqrt{2}}}{2}, \frac{\sqrt{2-\sqrt{2}}}{2}, i\right)$ i is $\mathbb{Z}_2$ + the other 2 are $\mathbb{Z}_4$ so $\mathbb{Z}_4 \times \mathbb{Z}_2$

9 roots: $\pm\sqrt{3}, \pm i\sqrt{2}$ so $\mathbb{Z}_2 \times \mathbb{Z}_2$

Problem 5

(5) 1 roots: $\pm 1, \pm i$ so primitive elem is i

2. roots: $\pm\sqrt{3}, \pm\sqrt{5}$ so $\mathbb{Q}(\sqrt{3}, \sqrt{5})$

WTS $\mathbb{Q}(\sqrt{3}, \sqrt{5}) = \mathbb{Q}(\sqrt{3}+\sqrt{5})$

$\supseteq$ is obv

$\frac{1}{\sqrt{3}+\sqrt{5}} = \frac{\sqrt{3}+\sqrt{5}}{(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})} = \frac{\sqrt{5}-\sqrt{3}}{2} \Rightarrow \sqrt{5}-\sqrt{3} \in \mathbb{Q}(\sqrt{3}+\sqrt{5})$

$\Rightarrow \sqrt{5}, \sqrt{3} \in \mathbb{Q}(\sqrt{3}+\sqrt{5})$ so $\subseteq$ $\therefore$

$\sqrt{3}+\sqrt{5}$ is primative elem

3 roots $\pm\sqrt{5}, \pm i\sqrt{3} \Rightarrow \mathbb{Q}(\sqrt{5}, i\sqrt{3})$ WTS $= \mathbb{Q}(\sqrt{5}+i\sqrt{3})$

$\supseteq$ is obv $\frac{1}{\sqrt{5}+i\sqrt{3}} = \frac{\sqrt{5}-i\sqrt{3}}{(\sqrt{5}+i\sqrt{3})(\sqrt{5}-i\sqrt{3})} = \frac{\sqrt{5}-i\sqrt{3}}{8} \Rightarrow \sqrt{5}-i\sqrt{3} \in$

So $\sqrt{5}, i\sqrt{3} \in \mathbb{Q}(\sqrt{5}+i\sqrt{3})$ so $\subseteq$ $\therefore$

$\sqrt{5}+i\sqrt{3}$ is prim elem

4 roots are $\sqrt[3]{2}$, $w\sqrt[3]{2}$ + $w^2\sqrt[3]{2}$ where $w = \frac{-1+i\sqrt{3}}{2}$

so $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$ b/c we need $\sqrt[3]{2} + i\sqrt{3}$ so

prim element is $\sqrt[3]{2}+i\sqrt{3}$ for same reasons as above

Problem 7

7 Let G be the Galois group. $G \leq S_3$

$|S_3| = 6$ so know $|G| \mid 6$ so $|G| = 1, 2, 3,$ or 6

Since $p(x)$ is of degree 3. If $F(\alpha_1)$ is a field extention, then we know $[F(\alpha_1):F] = 3$ so

$3 \mid |G|$ so $|G|$ is 3 or 6. If $|G|=3$ then

$G \cong \mathbb{Z}_3$, if $|G|=6$ then $G = S_3$

Problem 10

Professor said we can skip the questions about solvable

Problem 17

17 Let $ax^2+bx+c=0$ $\cancel{4a^2x^2+4abx+4ac=0}$

we can use quadratic formula to get

$r_1 = \frac{-b+\sqrt{\alpha}}{2a}$ $r_2 = \frac{-b-\sqrt{\alpha}}{2a}$ WTS $F(\sqrt{\alpha}) = F(r_1, r_2)$

$\supseteq$ is obvious. Since $\alpha \in F(\alpha)$ & $-b, 2a \in F$,

$\frac{-b \pm \sqrt{\alpha}}{2a} \in F(\sqrt{\alpha})$

$\subseteq$ $r_1 - r_2 = \frac{2\sqrt{\alpha}}{2a} = \frac{\sqrt{\alpha}}{a}$ so $ar_1 - ar_2 = \sqrt{\alpha}$ so $\sqrt{\alpha} \in F(r_1, r_2)$

$\therefore F(\sqrt{\alpha}) = F(r_1, r_2)$

Possible b/c $chr(F) = p$ so $2a \bmod p \not\equiv 0$

b/c p does not div 2 & p does not div a

(if $p \mid a$, $a=0$ so not quadiatic). so no div by 0